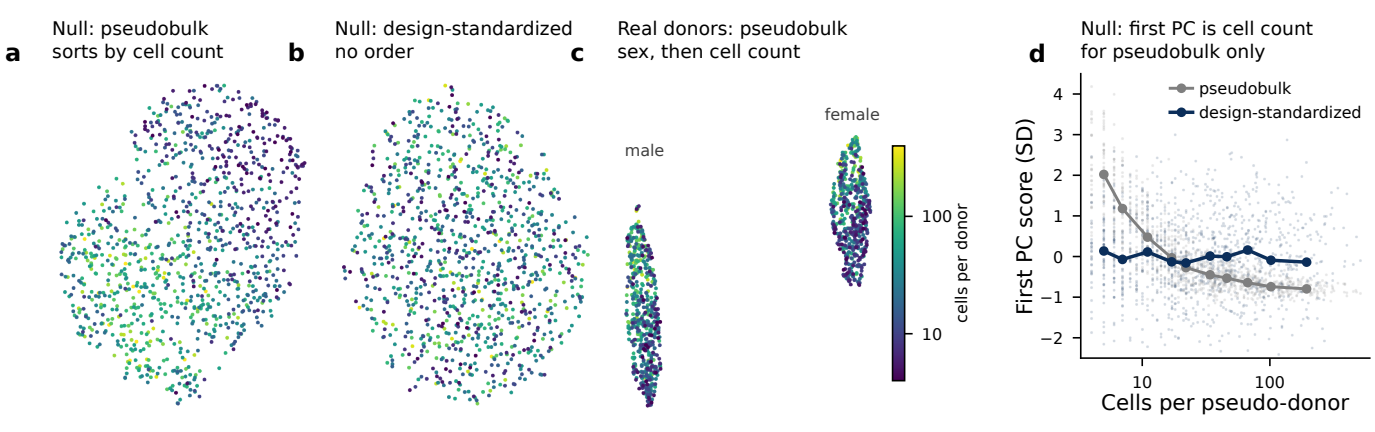

a
Null: pseudobulk
sorts by cell count
b
Null: design-standardized
no order
c
Real donors: pseudobulk
sex, then cell count
male
female
100
10
cells per donor
d
Null: first PC is cell count
for pseudobulk only
pseudobulk
design-standardized
First PC score (SD)
4
3
2
1
0
−1
−2
10
100
Cells per pseudo-donor